

What Should a 35-Year-Old Human Body Be Capable Of?

A practical benchmark report for movement, flexibility, body health, and heart health

Research draft for future health optimization page

June 29, 2026

Important Framing

This is a motivation and self-assessment document, not a medical diagnosis. The most useful question is not “am I elite?” but “which capability would give me the highest health return if I improved it by one band?” Benchmarks below combine official public-health guidance, clinical risk thresholds, and common field-test norms used in exercise science. Percentile labels are approximate because protocols, populations, equipment, and training history change the numbers.

How to Use This Report

Use the HTML page as the motivating dashboard and this PDF as the reference layer. The easiest path is:

1. Read the executive summary and public-health baseline.
2. Run the self-test battery once to get a starting point.
3. Pick one bottleneck: cardio, strength, mobility, balance, waist/body health, or recovery.
4. Follow the movement stack and P95+ progression for 8–12 weeks.
5. Retest using the same protocols, then choose the next bottleneck.

The report is organized from **why it matters** to **what good looks like** to **how to train and test it**. The HTML version should stay lighter: inspiration, next actions, and links back to this deeper research.

1 Executive Summary

For a healthy 35-year-old male or female, a robust movement baseline should include:

- **Cardio:** 150–300 min/week moderate aerobic work or 75–150 min/week vigorous work, plus the ability to brisk-walk for 30–45 minutes and complete occasional harder intervals.
- **Strength:** 2+ full-body strength sessions/week, enough upper-body strength for strict push-ups, enough lower-body strength for repeated squats/lunges, and enough grip strength to carry heavy groceries or luggage without strain.
- **Mobility:** painless squat, hip hinge, overhead reach, floor-to-stand, and basic hamstring/hip/ankle range.
- **Balance:** at least 30–45 seconds single-leg balance with eyes open; 10–20 seconds eyes closed is a useful stretch target.
- **Body and heart health:** normal blood pressure, waist-to-height ratio under about 0.5, resting heart rate generally 60–80 bpm, and recovery heart rate that drops quickly after exertion.

2 Core Public Health Baseline

Domain	Baseline target	Why it matters
Aerobic activity	150 min/week moderate, or 75 min/week vigorous, or a mix	CDC and U.S. guidelines associate this with broad cardiovascular, metabolic, mood, and longevity benefit.
Strength	2+ days/week, all major muscle groups	Preserves muscle, bone, glucose handling, joint resilience, and long-term independence.
Sedentary time	Break up long sitting periods	“Some physical activity is better than none”; frequent movement breaks help cardiometabolic risk.
Mobility	Daily exposure to squat, hinge, reach, rotate, and floor positions	Range that is not used tends to disappear; mobility supports pain-free activity choices.
Balance	Single-leg stance practice and loaded carries	Balance decline is quiet until it becomes expensive; train it before it becomes a fall-risk issue.

3 Approximate Capability Benchmarks at Age 35

Interpretation: **P25** means below-average but workable; **P50** is a practical median/average target; **P75** is a strong target; **P90** is aspirational for a recreationally fit person. Numbers are rounded.

3.1 Cardiorespiratory Fitness

Test	Sex	P25	P50	P75	P90 / strong
Estimated VO2 max, Male ml/kg/min	Male	35	40–42	45–48	50+
Estimated VO2 max, Female ml/kg/min	Female	24	28–31	34–36	38+
12-minute run distance	Male	1.8 km	2.1–2.3 km	2.5–2.7 km	2.8+ km
12-minute run distance	Female	1.5 km	1.8–2.0 km	2.2–2.5 km	2.6+ km
Resting heart rate	Any	80–90 bpm	65–75 bpm	55–65 bpm	45–55 bpm if well-trained
Heart-rate recovery, 1 min after hard effort	Any	12–17 bpm drop	18–25 bpm drop	26–35 bpm drop	35+ bpm drop

Practical goal: If you can do 30–45 minutes of brisk walking easily, one weekly Zone 2 session, and one brief interval session without feeling wrecked the next day, your heart-health foundation is moving in the right direction.

3.2 Muscular Strength and Endurance

Test	Sex	P25	P50	P75	P90 / strong
Strict push-ups, continuous	Male	12–16	17–24	25–34	35+
Strict push-ups, continuous	Female	8–11	12–19	20–26	27+
Forearm plank	Male	60 sec	90 sec	2–3 min	3+ min
Forearm plank	Female	45–60 sec	75–90 sec	2 min	3+ min
Grip strength, dominant hand	Male	40–43 kg	46–50 kg	52–56 kg	58+ kg
Grip strength, dominant hand	Female	24–26 kg	28–31 kg	33–36 kg	38+ kg
Bodyweight squat set	Any	15 clean reps	25 clean reps	40 clean reps	60+ clean reps

Practical goal: Be able to push, pull, hinge, squat, carry, brace, and get up from the floor. A very usable 35-year-old body does not need elite lifts; it needs reliable strength in ordinary human patterns.

3.3 Flexibility, Mobility, and Joint Capacity

Capability	Minimum	Strong target	Why it matters
Sit-and-reach	Male: 10–12 in; Female: 13–15 in	Male: 15–17+ in; Female: 18–19+ in	Hamstring/lower-back range; useful but not a full mobility picture.
Deep squat hold	30 sec, heels down or supported	2 min relaxed, heels down	Ankles, hips, knees, and trunk working together.
Knee-to-wall ankle test	5–8 cm from wall	10+ cm each side	Better squat, stair, running, and knee mechanics.
Shoulder overhead reach	Arms overhead without rib flare or pain	Full overhead flexion with loaded control	Supports lifting, reaching, swimming, climbing, and posture.
Floor-to-stand	Can get down/up without pain	Can rise without using hands	Whole-body coordination, strength, mobility, and balance.
Thoracic rotation	Rotate both ways without pain	45+ degrees each side	Helps shoulders, neck, breathing mechanics, and athletic movement.

Practical goal: Move through more positions than your chair asks of you. Flexibility is not just touching toes; it is keeping useful ranges available under control.

3.4 Balance and Coordination

Test	Baseline	Strong target	Notes
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Single-leg stance, eyes open	30 sec/side	45–60 sec/side	Should be quiet and controlled, not a fight.
Single-leg stance, eyes closed	5–10 sec/side	15–25 sec/side	Much harder; train near a wall.
Tandem walk	10 steps	20 controlled steps	Good simple coordination screen.
Loaded carry	30–60 sec	2+ min, moderate load	Combines grip, trunk, gait, and posture.

4 P95+ Compound Capability Targets

For this document, **P95+** means “stronger, fitter, and more mobile than roughly 95% of the general adult population,” not powerlifting, CrossFit, gymnastics, or endurance-sport elite status. These are high but realistic recreational targets for a 35-year-old who trains consistently for 1–3 years, has no major limiting injury, sleeps reasonably well, and progresses gradually.

4.1 The Minimum Compound Exercise Menu

To build a capable body, you do not need dozens of exercises. You need progressive exposure to a few human movement patterns:

- **Squat:** goblet squat, front squat, back squat, split squat.
- **Hinge:** Romanian deadlift, deadlift, kettlebell swing, hip thrust.
- **Push:** push-up, bench press, overhead press, dip.
- **Pull:** row, pull-up/chin-up, lat pulldown, loaded rope/towel pull.
- **Carry:** farmer carry, suitcase carry, front rack carry, backpack carry.
- **Trunk:** plank, side plank, dead bug, Pallof press, hanging knee raise.
- **Locomotion/conditioning:** brisk walking, running, cycling, rowing, swimming, burpees, stair climbing.

4.2 Bodyweight and Conditioning P95+ Targets

Exercise	P95+ target	Progression path
Burpees	50 clean reps in 5 minutes; aspirational: 100 in 10–12 minutes	Start with 5 sets of 5. Progress to 10 sets of 5, then 5 sets of 10, then timed 50. Keep reps clean: chest/floor standard optional, jump/stand standard consistent.
Strict push-ups	Male: 40–50; Female: 25–35	Incline push-up -¿ floor negatives -¿ floor push-ups -¿ deficit or weighted push-ups. Add reps only while body line stays clean.
Pull-ups/chin-ups	Male: 12–15; Female: 5–8	Dead hang -¿ scapular pulls -¿ band-assisted reps -¿ negatives -¿ strict reps -¿ weighted reps. Rows are the bridge if pull-ups are not ready.
Bodyweight squats	75–100 clean reps, or 50 reps in under 2 minutes	Squat to box -¿ full-depth squat -¿ tempo squats -¿ jump squats or loaded goblet squats.

Walking capacity	10 km comfortable walk; Daily steps -¿ 30-minute brisk walks 5 km brisk walk under 45 -¿ weekend long walk -¿ pace targets. minutes
Running capacity	5 km under 25 minutes for many men; under 28–30 minutes for many women Walk-run intervals -¿ 30-minute continuous easy run -¿ 1 tempo session/week -¿ 5 km test every 8–12 weeks.
Loaded carry	Carry 50–75% bodyweight total for 60–90 sec Grocery carry -¿ dumbbell farmer carry -¿ heavier carries -¿ longer distance or heavier load, not both at once.

4.3 Weight Training P95+ Targets

Use bodyweight-relative targets because absolute numbers are misleading. These are approximate P95+ general-population strength targets, not competitive lifting standards. A clean rep with full range and no pain beats a heavier ugly rep.

Lift	Male target	Female target	Notes
Back squat or front squat	1.5–1.75x bodyweight	1.0–1.25x bodyweight	Full-depth controlled squat; front squat target is usually lower than back squat.
Deadlift	2.0–2.25x bodyweight	1.5–1.75x bodyweight	Conventional, sumo, or trap-bar is fine; keep spine position consistent.
Bench press	1.25–1.5x bodyweight	0.65–0.85x bodyweight	Touch-and-press standard; no bouncing.
Overhead press	0.75–0.9x bodyweight	0.45–0.6x bodyweight	Standing strict press; no leg drive.
Pull-up/chin-up	12–15 strict reps or +25% bodyweight for 1 rep	5–8 strict reps or +10% bodyweight for 1 rep	Full hang to chin over bar, no kipping.
Farmer carry	0.75–1.0x bodyweight total for 60 sec	0.5–0.75x bodyweight total for 60 sec	Great practical strength test.

4.4 Minimum Strength Program to Reach Those Targets

Train strength 3 days/week. Two days/week can maintain and slowly build; three days/week is the better minimum for P95+ progress. Keep 1–3 reps in reserve on most sets, and test maxes rarely.

Day	Main work	Assistance
Day A	Squat 3–5 sets of 3–6 reps; bench press 3–5 sets of 3–8 reps	Row 3 sets, carry 3 rounds, trunk 2–3 sets
Day B	Deadlift 3–5 sets of 2–5 reps; overhead press 3–5 sets of 3–8 reps	Pull-ups or pulldowns 3–5 sets, split squats 3 sets

Day C	Front squat or Romanian deadlift 3–4 sets of 5–8 reps; incline/close-grip press 3–4 sets	Row/pull-up variation, carry, core, optional arms/calves
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Progression: start with a weight you can lift cleanly for all prescribed reps. Add 2.5–5 lb upper-body or 5–10 lb lower-body when all sets are completed with good form. If you miss reps two weeks in a row, reduce load 5–10% and rebuild. Every 4–8 weeks, take an easier week with fewer sets or lighter loads.

4.5 Burpee Progression Example

Phase	Target	Rule
Weeks 1–2	25 total reps, broken into sets of 5	Stop before form turns sloppy.
Weeks 3–4	50 total reps, broken into 5–10 sets	Reduce rest before increasing speed.
Weeks 5–8	50 reps for time	First goal: under 8 minutes. Next: under 6. Then: under 5.
Weeks 9–12	75–100 total reps occasionally	Use this as a test, not a daily workout. Protect wrists, shoulders, and low back.

4.6 Flexibility and Mobility P95+ Targets

P95+ flexibility is not contortion. It is owning large useful ranges without pain.

Capability	P95+ target	Progression
Deep squat	5 minutes relaxed, heels down, breathing easy	30 sec daily -¿ 2 min -¿ add prying, ankle rocks, and goblet counterweight.
Hamstring fold	Palms to floor or chest near thighs with soft knees allowed	Hip hinge drills -¿ strap hamstring stretch -¿ Jefferson curl with very light load if pain-free.
Ankle dorsiflexion	10–12+ cm knee-to-wall each side	Daily calf raises, bent-knee calf stretch, tibialis raises, loaded ankle rocks.
Hip rotation	Comfortable 90/90 switches without hands	90/90 holds -¿ active switches -¿ end-range lift-offs -¿ loaded split squats.
Shoulder overhead	Arms overhead with ribs down; loaded overhead press pain-free	Wall slides -¿ dead hangs -¿ thoracic extensions -¿ light overhead carries.
Floor-to-stand	Stand from floor without hands, both sides	Half-kneeling get-ups -¿ shin box get-ups -¿ Turkish get-up pattern.
Spinal rotation	45+ degrees thoracic rotation both sides	Open books -¿ quadruped rotations -¿ half-kneeling rotations -¿ loaded carries.

4.7 Stretching Extent: Useful Range Targets

Flexibility has a point of diminishing returns. The target is not maximum stretch; it is enough passive range, active control, and end-range strength to make normal life, lifting, sports, sitting, travel, and aging easier. If a range is painless, symmetrical, and usable under control, more range is optional.

Area	Functional baseline	Strong / P95+ use- ful range	How to test
Deep squat	30–60 sec with heels down or lightly supported	5 min relaxed, heels down, breathing through nose	Sit into squat; note heel lift, knee pain, back rounding, and whether breathing stays calm.
Ankle dorsiflexion	5–8 cm knee-to-wall each side	10–12+ cm each side, within 2 cm left/right	Knee touches wall while heel stays down and arch does not collapse.
Hamstrings	Sit-and-reach near toes or straight-leg raise 70–80°	Palms to floor with soft knees, or straight-leg raise 85–90°	Hinge from hips, not spine; avoid bouncing. A strap straight-leg raise is more specific.
Hip flexors	Half-kneeling lunge without low-back arch	Thigh extends behind body 10–20° with glute engaged	Posteriorly tilt pelvis; stretch front of hip, not low back.
Hip external rotation	Comfortable cross-legged sitting or 90/90 front shin	90/90 switches without hands; front hip near 90°	Compare sides; avoid knee pain.
Hip internal rotation	20–30° each side	35–45° each side, controlled	Seated or prone hip rotation; keep pelvis still. Often the missing hip range.
Shoulder flexion	Arms overhead without pain	170–180° overhead with ribs down and elbows straight	Back to wall; raise arms without flaring ribs or shrugging.
Shoulder external rotation	70–90° at 90° abduction	Symmetric 90° and comfortable loaded pressing/throwing prep	Test gently; do not force anterior shoulder stretch.
Thoracic rotation	35–45° each side	45–60° each side, smooth and symmetrical	Open-book or quadruped rotation; keep hips quiet.
Cervical rotation	60–70° each side	75–80° each side without pain	Turn head as if checking blind spot; no forcing.
Wrist extension	60–70°	80–90° for push-ups, handstands, front rack comfort	Palm on table/floor; progress slowly if wrists are sensitive.
Pancake / adductors	Comfortable wide straddle, upright torso	Chest moves forward in straddle without back pain	Train slowly; adductors dislike aggressive bouncing.

Floor-to-stand	Can rise with hand or knee help	Can rise without hands from both sides	Practical total-body mobility, balance, and strength screen.
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4.8 Stretching Progression Rules

- **Minimum dose:** 5–10 minutes daily beats one long weekly stretching session.
- **Intensity:** stretch at 3–5/10 discomfort, never sharp pain, numbness, tingling, or joint pinching.
- **Time:** hold static stretches 30–60 seconds for 1–3 rounds; use easy dynamic reps before training.
- **Active control:** after gaining range, add end-range lifts, loaded carries, slow squats, split squats, presses, or hinges so the body trusts the new range.
- **Symmetry:** a side-to-side gap matters more than a single absolute number; aim to keep most gaps small.
- **Hypermobility caution:** if you already exceed these ranges easily, prioritize strength, control, and joint stability over more stretching.

4.9 Stretching Standards by Outcome

Goal	Enough range	Useful practice
Pain-free daily life	Squat to pick things up, sit on floor, reach overhead, walk fast, climb stairs	Daily squat hold, hip flexor stretch, thoracic rotation, ankle rocks.
Strength training	Full-depth squat pattern, stable hinge, overhead reach, front rack or dumbbell rack tolerance	Mobility before lifting; loaded range after warm-up; avoid long passive stretching immediately before max strength.
Running / hiking	Ankle dorsiflexion 8–10+ cm, hip extension 10°+, calves resilient	Calf raises, tibialis raises, hip flexor work, hamstring flossing, step-down control.
Yoga / floor culture	Comfortable cross-legged, kneeling, deep squat, floor-to-stand	90/90 hip work, squat breathing, hamstring folds, shoulder flexion, gentle spinal rotation.
P95+ physical freedom	Most useful ranges are painless, symmetrical, and controlled; floor positions feel normal	10 min daily mobility plus 2–3 strength days that load the ranges.

4.10 Minimum Mobility Program

Do 10 minutes daily or 20 minutes 3–4x/week. Daily is better because mobility responds well to frequency.

1. 2 minutes deep squat hold or assisted squat hold.
2. 1 minute per side knee-to-wall ankle rocks.
3. 1 minute per side 90/90 hip switches or holds.
4. 1 minute per side hamstring hinge stretch.
5. 1 minute wall slides or dead hang.
6. 1 minute per side thoracic rotations.

4.11 A Realistic Timeline

- **0–3 months:** learn technique, build consistency, remove pain triggers, establish walking and mobility.
- **3–6 months:** visible strength and conditioning gains; first serious push-up, carry, squat, and cardio benchmarks.
- **6–12 months:** many people can reach above-average to strong status if training is consistent.
- **12–36 months:** P95+ strength and conditioning become realistic, especially for lifts and pull-ups.
- **Always:** sleep, protein, steps, and injury management decide whether progression compounds or stalls.

5 Body Health and Cardiometabolic Markers

Marker	Useful target	Notes
Blood pressure	Normal is below 120/80 mmHg	Repeated elevated readings deserve medical follow-up.
Waist circumference	Men below 40 in; women below 35 in	Simple proxy for visceral-fat risk, but height and ethnicity matter.
Waist-to-height ratio	Aim below 0.5	Often more intuitive than BMI for central fat.
BMI	18.5–24.9 is “normal” category	Useful at population level, imperfect for muscular people.
Body fat estimate	Men often healthiest around 12–22%; women around 21–33%	Method matters: DEXA, calipers, smart scales, and visual estimates can disagree.
Fasting glucose	Common clinical target below 100 mg/dL	Discuss persistent abnormal values with a clinician.
HbA1c	Common normal range below 5.7%	Useful for longer-term glucose exposure.
Sleep	7–9 hours/night	Movement progress gets easier when sleep is not fighting you.

6 A Motivating Movement Stack

This stack is designed to make progress feel doable rather than moralistic.

Minimum effective week

- 3 brisk walks of 30 minutes.
- 2 full-body strength sessions of 30–45 minutes.
- 5 minutes/day of mobility: squat, hinge, overhead reach, ankle rocks, thoracic rotation.
- 2–3 “movement snacks” per day: stairs, short walk, push-ups, squats, loaded carry, or mobility break.

Strong 35-year-old week

- 2 Zone 2 cardio sessions of 35–60 minutes.
- 1 short vigorous interval session.

- 2–3 strength sessions covering squat/lunge, hinge, push, pull, carry, and core.
- 10 minutes/day of mobility or tissue prep.
- One joyful physical activity: sport, hike, dance, swim, climb, long walk, bike ride.

7 Indian Cultural Defaults That Can Make Health Easier

Indian culture already contains many health-supporting defaults. The useful move is not to romanticize tradition or reject modern science; it is to keep the practices that reduce friction for better eating, movement, recovery, and community.

7.1 Food Decisions

Cultural pattern	Health-supporting use	Watch-outs
Dal, chana, rajma, sprouts, curd, paneer, eggs, fish, or lean meat	Build each meal around a visible protein source; this supports muscle, satiety, and glucose control.	Many vegetarian meals become mostly rice/roti plus potato unless protein is deliberately planned.
Thali structure	Use the plate as a dashboard: half vegetables/salad, one quarter protein, one quarter grain/starch, plus curd or dal.	Restaurant thalis often overload refined carbs, fried items, and sweets.
Home-cooked sabzi	Keep 2–3 vegetables in daily rotation; use spices for flavor instead of relying on sugar or excess oil.	Oil can quietly become the main calorie source; measure for a week to calibrate.
Fermented foods: idli, dosa, dhokla, kanji, curd, pickles	Good for variety, enjoyment, and gut-friendly patterns when paired with protein and fiber.	Fermentation does not automatically make a meal balanced; chutney, sambar, eggs, paneer, or dal help.
Masala chai ritual	Keep the ritual, reduce the sugar; pair with a walk, sunlight, or a protein snack.	Multiple sweet teas per day can become a hidden sugar habit.
Seasonal produce	Eat with seasons: gourds, greens, mango in moderation, citrus, millets, legumes, herbs.	“Natural” sweets and juices can still be high glucose loads. Whole fruit beats juice.
Fasting traditions	Use fasting as a mindful structure if it suits you: earlier dinner, no late-night snacking, simple foods.	Avoid turning fasts into fried/sugary feasts; diabetics, pregnant people, and those with medical conditions should get clinical advice.

7.2 Movement and Body Practice

- **Yoga as mobility plus breath:** use surya namaskar, malasana, lunges, twists, hip openers, and shoulder work as a daily 10–20 minute mobility base.
- **Walking after meals:** a 10–15 minute post-meal walk is culturally easy, socially acceptable, and useful for glucose control.
- **Floor culture:** sitting on the floor, squatting, and getting up without hands preserve hip, knee, ankle, and trunk capacity.

- **Household movement:** sweeping, mopping, carrying groceries, stairs, gardening, and cooking are not “fake exercise”; count them as movement snacks, then still add strength training.
- **Traditional strength tools:** gada/mace, clubs, dand-baithak, akhara-style bodyweight work, and loaded carries can be excellent when progressed safely.
- **Dance and sport:** garba, bhangra, classical dance, badminton, cricket, football, swimming, and hiking create joyful cardio without the psychology of punishment.

7.3 Daily Rhythm and Recovery

- **Earlier dinner:** many Indian households can shift dinner earlier than Western late-night patterns; this supports sleep and reduces late snacking.
- **Morning light:** chai, prayer, walking, or terrace time can double as circadian anchoring if done outdoors.
- **Community accountability:** walking groups, family step challenges, yoga classes, and sport meetups make health social rather than purely willpower-based.
- **Ritual as habit glue:** attach mobility to bath/prayer, walking to meals, strength to fixed weekdays, and weekly review to Sunday planning.

7.4 Indian Risk Context to Respect

South Asians often develop insulin resistance, type 2 diabetes, and central adiposity at lower BMI thresholds than many Western populations. That makes waist circumference, waist-to-height ratio, strength training, post-meal walking, and protein/fiber quality especially important. For a South Asian 35-year-old, the goal is not just weight control; it is protecting muscle, glucose handling, sleep, and waist size early.

7.5 Simple Household Defaults

- Keep roasted chana, sprouts, curd, eggs, paneer/tofu, dal, and fruit easy to reach.
- Make sweets and fried snacks occasional and intentional, not default pantry visibility.
- Serve salad/vegetables first, protein second, starch third.
- Prefer whole fruit over juice; prefer chaas or water over sweet drinks.
- Keep walking shoes visible near the door and use every meal as a cue for a short walk.
- Do a 5-minute floor routine daily: squat hold, hip switch, hamstring stretch, thoracic rotation, deep breathing.

8 Self-Test Battery

Repeat every 8–12 weeks. Do not test everything on the same day if you are new to exercise.

1. Resting blood pressure and resting heart rate.
2. Waist circumference and waist-to-height ratio.
3. 12-minute walk/run or a 1.5-mile run/walk.
4. One-minute heart-rate recovery after a hard but safe effort.
5. Strict push-ups to technical failure.
6. Forearm plank time.
7. Grip strength if a dynamometer is available; otherwise use loaded carry time.

8. Deep squat hold, sit-and-reach, knee-to-wall ankle test, and floor-to-stand.
9. Single-leg balance, eyes open and optionally eyes closed.

9 How to Test Every Movement Baseline

The goal is repeatability, not heroics. Test under similar conditions each time: same time of day, similar sleep, similar caffeine, same shoes or barefoot state, same surface, and no hard workout in the previous 24 hours. Stop any test if you feel chest pain, dizziness, sharp joint pain, unusual shortness of breath, or neurological symptoms.

9.1 Simple Equipment

- Stopwatch or phone timer.
- Measuring tape.
- Wall, chair, and clear floor space.
- Optional: heart-rate monitor, blood pressure cuff, grip dynamometer, yoga mat.
- Optional: a box/bench for step-ups and a safe outdoor/track route for walk/run testing.

9.2 Warm-Up Before Testing

1. Walk 5 minutes at an easy pace.
2. Do 10 bodyweight squats, 10 hip hinges, 10 arm circles each direction, and 10 ankle rocks each side.
3. Do 2 easy practice reps of any test that is unfamiliar.
4. Rest 2–3 minutes before starting the first scored test.

9.3 Recommended Test Order

1. Resting measures: resting heart rate, blood pressure, waist.
2. Mobility: squat, ankle, shoulder, thoracic rotation, floor-to-stand.
3. Balance: single-leg stance.
4. Strength endurance: push-ups, plank, squat set, loaded carry.
5. Cardio: 12-minute walk/run or step test last.

9.4 Resting Measures

Test	Protocol	Record
Resting heart rate	Measure after sitting quietly for 5 minutes, ideally before caffeine.	Beats per minute and context: sleep, caffeine, stress.
Blood pressure	Sit quietly, feet flat, arm supported. Take 2 readings 1 minute apart.	Average systolic/diastolic reading.
Waist circumference	Measure at navel level after normal exhale, tape snug but not compressing skin.	Inches or cm; also calculate waist divided by height.

9.5 Mobility Tests

Test	Protocol	Pass / score
Deep squat hold	Stand feet shoulder-width. Squat as low as comfortable. Try to keep heels down, knees tracking toes, chest relaxed. Hold.	Score time and quality: heels down, pain-free, relaxed breathing. Baseline: 30 sec. Strong: 2 min.
Knee-to-wall ankle test	Face wall. Put toes a measured distance from wall. Drive knee to wall without heel lifting or arch collapsing. Test both sides.	Record max distance in cm. Baseline: 5–8 cm. Strong: 10+ cm.
Sit-and-reach	Sit with legs straight, feet against wall or box. Reach forward slowly without bouncing.	Record farthest comfortable reach. Compare to your own prior result if no box scale.
Shoulder reach	overhead Stand with back against wall, ribs down, low back neutral. Raise both arms overhead.	Pass if arms reach near wall without pain, rib flare, or elbow bend. Record left/right restrictions.
Thoracic rotation	Sit tall or kneel on all fours. Rotate upper back left and right without forcing low back.	Pass if both sides feel smooth and roughly symmetrical. Note side difference.
Floor-to-stand	Sit on floor. Stand up with as little hand/knee assistance as safely possible.	Score number of supports used: hands, knees, furniture. Strong: no hands.

9.6 Balance Tests

Test	Protocol	Score
Single-leg stance, eyes open	Stand near wall. Lift one foot. Keep hips level and avoid gripping the floor aggressively. Stop at 60 sec.	Best of 2 attempts each side. Baseline: 30 sec. Strong: 45–60 sec.
Single-leg stance, eyes closed	Same setup, but close eyes after stable. Keep wall nearby. Stop at 30 sec.	Best of 2 attempts each side. Baseline: 5–10 sec. Strong: 15–25 sec.
Tandem walk	Walk heel-to-toe in a straight line. Arms can be out for balance.	Count controlled steps without stepping out. Baseline: 10. Strong: 20.

9.7 Strength and Muscular Endurance Tests

Test	Protocol	Score
Strict push-ups	Hands under shoulders, body straight, chest approaches floor, elbows lock out at top. Stop when form breaks. Use the same variation every retest.	Reps. If strict floor push-ups are not available, use incline push-ups and record surface height.

Forearm plank	Elbows under shoulders, glutes lightly squeezed, ribs down, neutral neck. Stop when hips sag, pain appears, or form breaks.	Time. Baseline: 45–90 sec. Strong: 2–3 min.
Bodyweight squat set	Feet comfortable width. Squat to a repeatable depth without knee pain or torso collapse. Stop at form breakdown.	Clean reps. Baseline: 15–25. Strong: 40+.
Loaded carry	Carry two equal weights, one in each hand, with tall posture. Walk slowly and safely.	Load, time, and distance. If no weights, use grocery bags or backpack.
Grip strength	Use a dynamometer if available. Arm by side, elbow about 90 degrees, squeeze hard for 3 sec.	Best of 3 each hand. Use same device/settings each time.

9.8 Cardio Tests

Test	Protocol	Score
12-minute walk/run	On a flat route or track, cover as much distance as possible in 12 minutes. Pace hard but sustainable.	Distance, average heart rate if available, and perceived effort 1–10.
30-minute brisk walk	Walk at the fastest pace you can sustain while still speaking in short sentences.	Distance, heart rate if available, and how recovered you feel after 10 minutes.
3-minute step test	Step up/down on a stable 12-inch step at a steady cadence for 3 minutes. Sit immediately and measure heart rate for 60 seconds.	Recovery heart rate. Lower is generally better, but compare mostly against yourself.
Heart-rate recovery	After a hard but safe effort, note peak heart rate, then stand or sit quietly for 1 minute.	Drop in bpm after 1 minute. Baseline: 12–17. Strong: 26+.

9.9 Recording Template

Metric	Today	Last time	Notes
Resting HR			Sleep, caffeine, stress
Waist-to-height			Same tape location
Deep squat hold			Heels down? pain?
Knee-to-wall L/R			Side difference
Single-leg balance L/R			Eyes open/closed
Push-ups			Strict or incline height
Plank			Form quality
Loaded carry			Load, distance, grip
12-minute distance			Route/weather/effort
Heart-rate recovery			Peak HR and 1-min drop

9.10 How to Interpret Results

- If cardio is low: start with walking frequency before intensity.
- If strength is low: prioritize 2 full-body strength sessions/week.
- If mobility is low: do 5–10 minutes daily; frequency beats occasional long stretching.
- If balance is low: practice near a wall for 2 minutes/day.
- If waist or blood pressure is high: combine walking, strength, sleep, and food defaults; consider medical follow-up for repeated abnormal readings.
- If one side is much worse than the other: train both sides, but give the weaker side extra gentle volume.

10 Translation Notes for the HTML Page

Reviewed page: <https://priyankajayaswal.github.io/til/healthoptimization.html>

The existing page is strong on mechanisms and interventions: visceral fat, sleep, nutrition and fasting, supplements, toxins, brain health, hormones, immune function, GLP-1 drugs, peak span, recommendations, and a fun exercise tab. This research adds the **capability layer**: concrete tests, sex-specific benchmarks, P95+ targets, stretching ranges, and a repeatable journey from baseline to higher capacity.

Recommended HTML treatment

- Keep the Capability tab motivational and action-oriented; avoid turning the page into the full PDF.
- Use the PDF as the detailed reference for exact numbers, protocols, and caveats.
- Lead with physical freedom and peak span: staying close to your best function for longer.
- Make the first action obvious: test once, choose one bottleneck, follow an 8–12 week plan, retest.
- Keep Indian cultural defaults as practical friction-reducers: thali structure, post-meal walks, floor culture, visible protein, and family/community accountability.
- Add future interactivity only if it reduces decision-making: saved baselines, percentile bands, next-cycle recommendations, and printable test sheets.

11 Research Notes and Sources

- CDC adult activity guidance: <https://www.cdc.gov/physical-activity-basics/guidelines/adults.html>
- U.S. Physical Activity Guidelines for Americans: <https://odphp.health.gov/our-work/nutrition-physical-activity/physical-activity-guidelines/current-guidelines>
- American Heart Association blood pressure categories: <https://www.heart.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-readings>
- American Heart Association target heart rate guidance: <https://www.heart.org/en/healthy-living/fitness/fitness-basics/target-heart-rates>
- CDC healthy weight and BMI background: <https://www.cdc.gov/bmi/adult-calculator/bmi-categories.html>
- WHO expert consultation on BMI and risk in Asian populations: <https://pubmed.ncbi.nlm.nih.gov/14726171/>
- Diabetes risk in Asian Indian and South Asian populations is often elevated at lower BMI and waist levels; use ethnicity-aware clinical guidance where available.
- U.S. Physical Activity Guidelines emphasize gradual progression, safe activity choices, 150–300 minutes/week of moderate aerobic work or 75–150 minutes/week vigorous work, and 2+ days/week of muscle-strengthening for all major muscle groups.

- P95+ strength targets are practical recreational targets synthesized from common strength-standard tables and coaching practice; verify against a chosen standard before publishing exact percentile claims.
- Joint range-of-motion values are rounded from common goniometry references such as AAOS joint-motion measurement conventions and clinical texts such as Norkin and White’s *Measurement of Joint Motion*. Individual anatomy, sport, injury history, and hypermobility status change what is desirable.
- Stretching targets emphasize useful active range, symmetry, and pain-free control rather than maximum passive flexibility.
- Field-test norms are rounded from commonly used ACSM/YMCA/Cooper-style exercise testing categories and should be replaced with a fully cited table before public publication.
- Grip-strength estimates are rounded from adult dynamometry reference norms; use the same device and protocol over time if tracking yourself.

Next Version Ideas

- Add charts for male/female P25/P50/P75/P90 capability bands.
- Add an interactive calculator: age, sex, test result, percentile band, next habit.
- Add a “choose your bottleneck” flow: cardio, strength, mobility, balance, body composition, or recovery.
- Add an India-specific version of the calculator with vegetarian protein defaults, South Asian waist-risk guidance, and culturally familiar movement options.
- Add citations for each exact norm table before publishing publicly.